DATA SCIENCE USING PYTHON - MAT337

CIA – I

1. A data science team is performing preliminary processing on a set of research data before applying advanced analytical techniques. Develop suitable Python programs for the following requirements:

a) A researcher records n sequential observations during an experiment. Write a Python program to determine the total value represented by all observations from 1 to n.

b) During data preprocessing, two numerical values stored in variables A and B are assigned to the wrong fields. Correct the assignment by exchanging their values without creating another variable.

c) Three sensors provide numerical readings for the same parameter. Write a Python program to determine which reading represents the highest observed value.

d) A dataset contains a text-based identification code. For a validation step, the analyst needs to view the code in the opposite order. Write a Python program to generate the required representation.

e) A numerical record ID is received by the system. Write a Python program to determine how many individual digits are present in the ID. The program should work correctly even when a negative integer is entered.

2. A college is conducting a "Fun with Numbers and Words" activity where students participate in different small challenges involving their marks, numbers, and names. The organizer wants to use Python to automate the checking and calculation process. Develop suitable Python programs for the following activities:

a) The organiser has collected the marks of a group of students in a surprise test. Calculate a single value that represents the overall performance of the group.

b) Each participant is given a number at the entry counter. Based on the nature of the number, place the participant into one of two groups.

c) A participant claims that their chosen nickname reads exactly the same from both directions. Write a program to verify whether the claim is true.

d) During the final round, a participant gives a number and claims that it has a special mathematical property. Verify the claim using the digits of the number and the total number of digits.

3. An online shopping company is developing a Python-based system to manage customer orders and product information. The system receives customer names, product IDs, customer reviews, product prices, and sales information. Since some information may be repeated or require analysis, the company wants to automate these tasks using Python. The system should analyze customer names, remove duplicate product IDs, identify frequently used words in customer reviews, calculate total prices, and find the highest-priced product.

a) Write a Python program to accept a customer's name and count the number of vowels present in the name.

b) Write a Python program to accept a list of product IDs and remove duplicate product IDs.

c) Write a Python program using a dictionary to find the frequency of each word in a customer review.

d) Define a user-defined function to accept the prices of two products and return their total price.

e) Define a user-defined function to accept a list of product prices and return the highest-priced product.

4. A wildlife department maintains records of animals found in different forest areas. The system stores information such as animal identification numbers, population counts, survey intervals, and identification codes. The department wants to use Python programs to analyze these records and obtain useful information.

Answer the following questions based on the above case study:

a) Two animals are observed at intervals of 12 days and 18 days. Find the greatest number of days that can be used as a common observation interval.

b) A forest record contains a number whose factors, excluding the number itself, add up to the same number. Write a program to check whether the given number has this property.

c) The population counts of animals in different areas are given as a list. Find the second-highest population count without sorting the list.

d) An animal identification code contains letters, numbers, and symbols. Count the number of uppercase letters, lowercase letters, digits, and special characters in the code.

5. A bank is developing a Digital Banking System to manage customer accounts, transactions, and security information. The system receives numerical transaction values, customer messages,

account details, and passwords. The bank wants to process this information automatically and identify important details from the data.

Answer the following based on the above case study:

a) A transaction value is stored by the system in decimal form, but different banking systems require different number representations. Display the value in the three commonly used representations.

b) A customer sends a transaction message to the bank. The system needs to identify the first character that occurs only once in the message.

c) Transaction data is stored in rows and columns for further processing. Convert the given arrangement so that its rows become columns and columns become rows.

d) Before allowing a customer to create an account password, the system needs to verify that it satisfies the bank's basic security requirements.

e) The bank stores customer names along with their account balances. Identify the customer who currently has the highest balance.

6. A software development team is developing a Python-based system to process and organise information collected during a college event. As part of the data-processing stage, the team needs to perform several operations on lists containing scores, participant records, activity details, and item information. Develop suitable Python programs for the following requirements:

a) During an event, the same score may be recorded more than once. The system should retain only one occurrence of each score and present the available scores from the smallest to the largest. Implement this requirement using the given list of scores.

b) A participant's scores from different evaluation rounds are stored in a list. To avoid considering the two extreme scores, the smallest and largest scores are to be excluded. The remaining scores should then be displayed starting from the last score to the first. Implement this requirement using the given list.

c) Two activity coordinators maintain separate lists of participant identification numbers. Some participants may have attended both activities. Determine the identification numbers that appear in both lists and present them in increasing order.

d) The event organisers maintain a list of food items ordered for the event. The item "apple" may appear several times in the list. Find how many times it was ordered and then clear all occurrences of "apple" from the list. Display the number of apples ordered along with the updated list.

e) A stall maintains a list of item prices. A new item priced at 25 needs to be placed at the third position, while the item priced at 10 is no longer available. Update the price list accordingly and display the resulting list.

7. A library management system is being developed to organise book codes and borrowing records. The librarian needs to perform the following operations on the stored data:

a) A collection contains repeated book codes, and two new books with codes 4 and 5 are added. The book with code 2 is later removed from the collection. Obtain the final collection without retaining duplicate codes.

b) A librarian wants to prepare a highest-to-lowest arrangement of book numbers for reference, but the original order must remain unchanged. Obtain both the original list and the newly arranged list.

c) A student's borrowed book codes are stored in a tuple. Find the position of the first occurrence of book code 4 and determine how many times it appears.

d) Two students have borrowed books represented by two sets. The librarian wants the books that belong only to the first student's record, excluding those also found in the second student's record. Obtain the updated first set.

e) Three students have borrowed books from the library, with their book codes stored separately. Identify the book code(s) that were borrowed by every one of the three students.

8. A software team is developing a Python-based data analysis system to identify useful patterns in numerical records and text codes. Develop suitable Python programs for the following requirements:

a) A researcher needs to examine a sequence of recorded values and determine the longest continuous period during which the observations keep increasing from one record to the next.

b) After identifying such a period, the system should keep track of the current increasing length and update the maximum length whenever a longer sequence is found.

c) A communication system receives two text codes and needs to determine whether one code can be formed from the other by moving some characters from the beginning to the end.

d) Before comparing the two codes, the system should ensure that they contain the same number of characters and then provide a suitable result indicating whether the required relationship exists.

e) A data analyst is examining frequently occurring values in a dataset. The system should identify the values whose frequency is greater than one-third of the total number of observations, while ensuring that the same value is reported only once.

9. A software company is developing a text and numerical data processing system. The system is required to analyse text patterns, numerical values, and words to obtain useful information. Develop suitable Python programs for the following requirements:

a) A text-processing system needs to identify the longest sequence of characters at the beginning of a given text that also appears at its end.

b) The system should examine a collection of numerical values and determine the largest possible product that can be obtained by selecting any three values from the collection.

c) A number is stored as a sequence of digits. The system allows the user to remove exactly one digit and wants to obtain the largest possible number after the removal.

d) A sentence contains several words. The system needs to determine which word contains the greatest number of characters that occur more than once within that word.

e) For the selected word, the system should also determine the number of characters that are repeated and use this information to identify the word with the highest repetition among all the words in the sentence.

10. A warehouse manager is reviewing product arrangements, identification codes, and daily price records to improve inventory management. Develop suitable Python programs for the following requirements:

a) Products are arranged according to their identification numbers, but some of them are in the wrong positions. Rearrange them into the required order and determine the number of exchanges needed.

b) Before making any changes to the product arrangement, the manager wants to preserve the original arrangement for later reference.

c) Some product identification codes contain groups of characters that read the same from both directions. Identify all such groups containing more than one character.

d) The selling price of a product changes over several days. Determine the largest increase that occurred between an earlier day and a later day.

e) While analysing the price changes, the manager wants to compare each later price with the lowest price recorded before it and identify the maximum possible increase.